

1. Basics of Data Science

Q1. What is Data Science?

A: Data Science is an interdisciplinary field that uses statistical methods, algorithms, and tools to extract knowledge and insights from structured and unstructured data.

Q2. What is Data Analytics?

A: It is the process of examining data sets to draw conclusions and support decision-making.

Q3. Difference between Data Science and Big Data?

A: Data Science focuses on analysis and insights, while Big Data focuses on handling and processing large-scale data.

Q4. What is a dataset?

A: A collection of related data organized in tabular or other formats.

Q5. What are features and labels?

A: Features are input variables; labels are output variables.

Q.6.Explain the Data Science lifecycle.

Answer:

The Data Science lifecycle consists of several stages:

1. **Data Collection** – Gathering data from databases, sensors, APIs, etc.
2. **Data Cleaning** – Removing errors, duplicates, and missing values.
3. **Data Exploration** – Understanding patterns using statistics and visualization.
4. **Feature Engineering** – Selecting important features for modeling.
5. **Model Building** – Applying machine learning algorithms.
6. **Model Evaluation** – Checking accuracy and performance.
7. **Deployment** – Using the model in real-world applications.
8. **Monitoring** – Updating and maintaining model performance.

Q.7.What is Exploratory Data Analysis (EDA)?

Answer:

EDA is the process of analyzing datasets using statistical methods and visualization techniques to understand patterns, relationships, and anomalies before applying machine learning algorithms.

Q.8.What is feature engineering?

Answer:

Feature engineering is the process of selecting, modifying, or creating new features from raw data to improve machine learning model performance.

Example:

Creating “Age Group” from “Age”.

■ 2. Python & Libraries

Q9. What is NumPy?

A: A library used for numerical computations with arrays and matrices.

Q10. What is Pandas?

A: A library used for data manipulation and analysis using DataFrames.

Q11. What is a DataFrame?

A: A 2D labeled data structure with rows and columns.

Q12. What is Matplotlib?

A: A library used for data visualization.

Q13. What is Scikit-learn?

A: A library used for implementing machine learning algorithms.

Q14. Why is Python preferred for Data Science?

Answer:

Python is preferred because:

- Easy syntax
- Large community support
- Rich libraries for ML and visualization
- Platform independent
- Supports AI and Big Data tools

Q15. Explain important Python libraries used in Data Science.

Answer:

Library	Purpose
NumPy	Numerical operations
Pandas	Data manipulation
Matplotlib	Visualization
Seaborn	Statistical plots
Scikit-learn	Machine learning
TensorFlow	Deep learning

Q16 What is vectorization in NumPy?

Answer:

Vectorization means performing operations on arrays without using loops, which improves execution speed.

Example:

```
import numpy as np  
a = np.array([1,2,3])  
print(a*2)
```

3. Data Preprocessing

Q17. What is data preprocessing?

A: The process of cleaning and transforming raw data.

Q18. Why is preprocessing important?

A: It improves data quality and model accuracy.

Q19. What are missing values?

A: Data points that are not available in a dataset.

Q20. How do you handle missing values?

A: By removing or filling them using mean, median, or mode.

Q21. What is normalization?

A: Scaling data between 0 and 1.

Q22. What is standardization?

A: Scaling data to mean = 0 and standard deviation = 1.

Q23. What is encoding?

A: Converting categorical data into numerical form.

Q24. Why is data preprocessing necessary?

Answer:

Real-world data is often incomplete, noisy, and inconsistent. Preprocessing improves data quality and model accuracy.

Benefits:

- Removes errors
- Handles missing values
- Reduces redundancy
- Improves efficiency

Q25 What are outliers?

Answer:

Outliers are extreme values significantly different from other observations.

Example:

Marks: 45, 50, 48, 47, 500

Detection Methods:

- Boxplot
- Z-score
- IQR method

Q26 Explain normalization and standardization.

Answer:

Normalization

Standardization

Scales data between 0 and 1 Converts data to mean 0 and std 1

Used when range matters Used for normally distributed data

Q27 What is one-hot encoding?

Answer:

One-hot encoding converts categorical values into binary columns.

Example:

Color

Red

Blue

Becomes:

Red Blue

1 0

0 1

4. Data Visualization

Q28. What is data visualization?

A: Representation of data in graphical form.

Q29. What is a histogram?

A: A graph showing frequency distribution.

Q30. What is a scatter plot?

A: Shows relationship between two variables.

Q31. What is a boxplot?

A: Displays data distribution and outliers.

Q32 Why is visualization important in Data Science?

Answer:

Visualization helps understand trends, patterns, relationships, and anomalies in data easily.

Q33 Difference between bar chart and histogram?

Answer:

Bar Chart	Histogram
Used for categorical data	Used for continuous data
Bars separated	Bars connected

Q34 What is correlation?

Answer:

Correlation measures the relationship between two variables.

Types:

- Positive correlation
 - Negative correlation
 - Zero correlation
-

5. Machine Learning Basics

Q35. What is Machine Learning?

A: A method where machines learn patterns from data.

Q36. Types of Machine Learning?

A: Supervised, Unsupervised, Reinforcement.

Q37. What is overfitting?

A: When model performs well on training data but poorly on new data.

Q38. What is underfitting?

A: When model fails to capture patterns in data.

Q39 What is supervised learning?

Answer:

A type of ML where the model learns using labeled data.

Examples:

- Regression
- Classification

Q40 What is unsupervised learning?

Answer:

Learning from unlabeled data.

Examples:

- Clustering
- Association rules

Q41. Explain reinforcement learning.

Answer:

A learning method where an agent learns by interacting with environment and receiving rewards or penalties.

Example:

Robot navigation, game AI

Q42. What is bias and variance?

Answer:

Bias

Variance

Error due to assumptions Error due to sensitivity

Causes underfitting

Causes overfitting

Q43. What is cross-validation?

Answer:

A technique used to evaluate model performance using multiple train-test splits.

6. Regression

Q44. What is regression?

A: Technique to predict continuous values.

Q45. What is linear regression?

A: A model that fits a straight line to data.

Q46. What is MSE?

A: Mean Squared Error – average squared difference between actual and predicted values.

Q47. What is R² score?

A: Indicates how well data fits the model.

Q48. Explain Linear Regression mathematically.

Answer:

$$y=mx+c$$

Where:

- y = dependent variable
- x = independent variable
- m = slope
- c = intercept

It predicts continuous output values.

Q49. Assumptions of Linear Regression?

Answer:

- Linear relationship
- No multicollinearity
- Homoscedasticity
- Normal distribution of errors

Q50. What is gradient descent?

Answer:

An optimization algorithm used to minimize error by updating parameters iteratively.

7. Classification

Q51. What is classification?

A: Predicting categorical outcomes.

Q52. What is Logistic Regression?

A: A classification algorithm used for binary outcomes.

A supervised learning algorithm used for classification problems.

Formula:

$$P(y) = \frac{e^{-x}}{1 + e^{-x}}$$

It predicts probabilities between 0 and 1.

Q53. What is accuracy?

A: Ratio of correct predictions to total predictions.

Q54. What is precision?

A: Correct positive predictions out of total predicted positives.

Q55. What is recall?

A: Correct positives out of actual positives.

Q56. What is F1-score?

A: Harmonic mean of precision and recall.

Q57 Explain confusion matrix.**Answer:**

	Predicted Positive	Predicted Negative
Actual Positive	TP	FN
Actual Negative	FP	TN

Used to evaluate classification models.

7.Algorithms

Q58. What is Naïve Bayes?

A: A probabilistic classifier based on Bayes theorem.

Q59. What is Decision Tree?

A: A tree-based model for decision making.

Q60. What is KNN?

A: Classifies based on nearest neighbors.

Q61. What is K-Means?

A: Clustering algorithm that groups data into k clusters.

Q62.What is clustering?

Answer:

Grouping similar data points together.

Q63 Explain K-Means algorithm.

Answer:

K-Means divides data into K clusters.

Steps:

1. Select K centroids
2. Assign points to nearest centroid
3. Update centroids
4. Repeat until convergence

8. Big Data Concepts

Q64. What is Big Data?

A: Data that is too large to process using traditional methods.

Q65. What are the 5 V's?

A: Volume, Velocity, Variety, Veracity, Value.

Q66. What is Hadoop?

A: A framework for distributed storage and processing.

Q67. What is HDFS?

A: Hadoop Distributed File System.

Q68. What is MapReduce?

A: Programming model for processing large datasets.

Q69. What is Apache Spark?

A: Fast in-memory data processing engine.

Q70 Explain Hadoop architecture.

Answer:

Hadoop architecture contains:

- HDFS → Storage
- MapReduce → Processing
- YARN → Resource management

Q71 Difference between Hadoop and Spark?

Answer:

Hadoop	Spark
Disk-based	Memory-based
Slower	Faster
Batch processing	Real-time processing

Q72 What is HDFS?**Answer:**

Hadoop Distributed File System stores large data across multiple machines.

Features:

- Fault tolerant
- Distributed storage
- High throughput

Q73 What is MapReduce?**Answer:**

A programming model for parallel data processing.

Phases:

1. Map phase
2. Shuffle phase
3. Reduce phase

9 Practical-Based Questions

Q74. Which dataset did you use?

A: (Student should explain their dataset)

Q75. What preprocessing steps did you apply?

A: Handling missing values, normalization, etc.

Q76. Which algorithm did you use and why?

A: Based on problem type (regression/classification).

Q77. How did you evaluate your model?

A: Using accuracy, MSE, etc.

Q78. How can you improve model performance?

A: Hyperparameter tuning, more data, feature selection.

Q79 What is Deep Learning?

Answer:

A subset of machine learning using neural networks with multiple hidden layers.

Q80 What is Neural Network?

Answer:

A computational model inspired by the human brain.

Components:

- Input layer
- Hidden layer
- Output layer

Q81 What is IoT?

Answer:

Internet of Things connects physical devices to the internet for communication and automation.

Q82. What is Cloud Computing?

Answer:

Delivery of computing services over the internet.

Types:

- IaaS
- PaaS
- SaaS

Q83 What is Cybersecurity?

Answer:

Protecting systems and data from cyber attacks.

10. Advanced Viva Questions

Q84. What is cross-validation?

A: Technique to validate model using multiple splits.

Q85. What is hyperparameter tuning?

A: Optimizing model parameters.

Q86. What is bias and variance?

A: Bias = error due to assumptions; Variance = sensitivity to data.

Q87. What is ROC curve?

A: Graph showing classification performance.

Q88. What is AUC?

A: Area under ROC curve.

Q89 Why did you select this dataset?

Answer:

Because it is relevant to the problem statement and contains sufficient data for analysis.

Which preprocessing techniques did you use?

Answer:

- Handling missing values
- Removing duplicates
- Normalization
- Encoding

Q90 Which evaluation metric did you use?

Answer:

- Accuracy for classification
- MSE/RMSE for regression

Q91 How can overfitting be reduced?

Answer:

- Cross-validation
- More training data
- Regularization
- Pruning

Q92 What are real-world applications of Data Science?

Answer:

- Healthcare prediction
- Fraud detection
- Recommendation systems
- Smart cities
- Autonomous vehicles